

SmartGym Adaptive Training System

Firestore Dashboard Seed Guide

Seed data structure and safe dashboard test imports.

Version 1.0 Prototype

ESP32-S3 firmware | Firestore RTDB | React dashboard

Generated documentation package

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Firestore Dashboard Seed Guide

This document explains the large dashboard seed file:

- [firebase_rtdb_dashboard_seed_2026_w12_w17.json](C:/Users/chomp/OneDrive/Viejo/Documentos/PlatformIO/Pr

It was generated by:

- [generate_dashboard_seed.py](C:/Users/chomp/OneDrive/Viejo/Documentos/PlatformIO/Projects/SmartGym_Displ

Purpose

This seed is meant to validate a full SmartGym dashboard, not just a simple demo.

It includes enough data to test:

- user list screens
- current device status
- weekly and daily activity views
- machine-specific progress
- rep and set detail pages
- body metric trends
- post-session feedback like 'RPE' and notes

Dataset Size

Current generated dataset:

- '5' users
- '5' machines
- '6' weeks
- '147' training days
- '673' machine sessions

Each user trains roughly '4-6' days per week.

Each training day contains '4-5' machine sessions.

Top-Level Nodes

The seed contains these main roots:

- 'usersByRfid'
- 'calibrations'
- 'calibrationHistory'

- 'machineConfigs'
- 'devices'
- 'bodyMetricsHistory'
- 'sessionFeedback'
- 'athleteWeeklySessions'

What Each Root Means

'usersByRfid'

This is the athlete profile table.

Each user has:

- RFID UID
- display name
- body stats
- primary goal

Use this for:

- athlete selector
- profile header
- goal-based filtering

'calibrations'

This is the current effective calibration per user and machine type.

Each record contains:

- 'suggestedWeightKg'
- 'userRomPercent'
- 'hasCalibration'

Use this for:

- readiness checks
- showing recommended weight
- machine onboarding logic

'calibrationHistory'

This is the historical calibration log.

Use this for:

- calibration trend charts

- seeing if suggested weight changed over time
- auditing recalibration events

‘machineConfigs‘

This is the machine metadata and recommendation source.

Each machine contains:

- machine IDs
- ROM defaults
- default calibration weight
- goal recommendations
- encoder config fields
- taxonomy fields:
- ‘exerciseCategory‘
- ‘primaryMuscleGroup‘
- ‘secondaryMuscleGroup‘

Use this for:

- machine cards
- exercise grouping
- muscle group dashboards
- settings/admin screens

‘devices‘

This is the live device state table.

Each device contains:

- device identity
- assigned machine
- active user UID
- app state
- wifi state
- encoder calibration state

Use this for:

- live gym-floor monitor
- ?which machine is active right now??
- device diagnostics

‘bodyMetricsHistory’

This is web-app-owned athlete check-in history.

Each date entry may contain:

- weight
- body fat
- skeletal muscle percentage
- waist
- sleep
- energy score
- notes

Use this for:

- transformation tracking
- bodyweight charts
- recovery dashboards

‘sessionFeedback’

This is web-app-owned post-workout feedback.

Each session may contain:

- ‘rpe’
- ‘energyScore’
- ‘sorenessScore’
- ‘painScore’
- ‘readinessScore’
- ‘notes’

Use this for:

- session review screens
- coach notes overlays
- combining objective and subjective progress

‘athleteWeeklySessions’

This is the main workout history tree.

Path shape:

```
athleteWeeklySessions/{uid}/{weekKey}/days/{dayKey}
```

Inside each day:

- 'meta'
- 'daySummary'
- 'timeline'
- 'sessions'

Use this for:

- weekly dashboard
- day timeline
- full session analysis

Session Structure

Each machine session lives here:

```
athleteWeeklySessions/{uid}/{weekKey}/days/{dayKey}/sessions/{sessionId}
```

A session includes:

- 'identity'
- 'machine'
- 'timing'
- 'plan'
- 'summary'
- 'analysis'
- 'setOverview'
- 'setDetails'
- 'repSets'
- 'representativeReps'

'summary'

This is the fast session result block.

Examples of fields:

- 'setsCompleted'
- 'validReps'
- 'invalidReps'
- 'volumeLoadKg'
- 'avgRomPercent'

- 'avgPeakVelocityPctPerSec'

'analysis'

This is the richer dashboard/coaching block.

Examples of fields:

- 'romComplianceRate'
- 'idealRomHitRate'
- 'avgRestSecondsPerSet'
- 'fatigueRomDrop'
- 'fatigueVelocityDrop'
- invalid reason counters

'setOverview'

This is the per-set summary array.

Use it for:

- set cards
- fatigue comparison
- quick coach review

'setDetails'

This is a more explicit set-by-set object.

Use it for:

- set tables
- deeper analysis views

'repSets'

This is the rep-level detail grouped by set.

Each rep can include:

- validity
- ROM
- duration
- concentric time
- peak concentric velocity
- peak eccentric velocity
- warning flags

Use it for:

- motion charts
- rep scatter plots
- coaching detail pages

How The Seed Was Built

The generator creates realistic but synthetic gym usage.

Main rules:

- users train '4-6' times per week
- each training day uses '4-5' machines
- sessions vary by goal, machine, and calibration
- reps include valid and invalid attempts
- some sessions include fatigue and warning patterns
- body metrics and feedback are included separately from device data

This means the seed is good for UI testing, filtering, and chart validation, but it is still synthetic data, not a physiological model.

Best Ways To Explore It

If you want to understand the file quickly, inspect it in this order:

1. 'usersByRfid'
2. 'machineConfigs'
3. one user under 'athleteWeeklySessions'
4. one week
5. one day
6. one session
7. that session?s 'summary', 'analysis', 'setDetails', and 'repSets'

That path gives you the whole mental model without needing to read the entire file.

Recommended Dashboard Tests

This seed is especially useful to validate:

1. Weekly activity charts
2. Session count and volume summaries
3. Machine filtering by type and muscle group
4. Body metrics trend charts

5. Session detail pages with rep analysis
6. Coach review panels using 'RPE' and notes
7. Device-monitor screens using 'devices'

Regenerating The Seed

To regenerate the seed:

```
@'
import runpy
runpy.run_path('tools/generate_dashboard_seed.py', run_name='__main__')
'@ | python -
```

The output file will be rewritten at:

- [firebase_rtdb_dashboard_seed_2026_w12_w17.json](C:/Users/chomp/OneDrive/Viejo/Documentos/PlatformIO/Pro

Important Note

This seed contains both:

- firmware-owned data
- machine taxonomy
- sessions
- calibrations
- devices
- web-app-owned data
- 'bodyMetricsHistory'
- 'sessionFeedback'

That split is intentional so your dashboard can validate the complete product flow, not only the embedded device upload.